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last updated: 2026-11-13

RESEARCH INTERESTS & PROFILE

- Postdoctoral research associate studying evolutionary genomics of cancer, with a focus on reconstructing cancer phylogeny at single-cell resolution.
- Methods developer for comparative sequence analysis: property-informed evolutionary models (PRIME) in the HyPhy/Datamonkey ecosystem.
- Interdisciplinary background spanning bioinformatics and human-computer interaction, with a research program in bioinformatics education and tutorials in the era of generative AI.
- Community builder experienced at initiating and facilitating collaborations across disciplines and career stages.

EDUCATION

PhD in Bioinformatics, Temple University Aug 2019 - Aug 2026
Advisor, Bioinformatics: Dr. Sergei L. Kosakovsky Pond
Co-Advisor, Human-Computer Interaction (HCI): Dr. Stephen MacNeil
Dissertation Topic: Harnessing interdisciplinarity in the investigation of nucleotide sequence evolution
[Recap | Abstract | Shorts | Dissertation]

MS in Computational Biology, Carnegie Mellon University Aug 2015 - Dec 2016
Relevant Coursework: Computational Genomics, Machine Learning, and Algorithms & Advanced Data Structures

BS in Chemistry, Carnegie Mellon University Aug 2010 - May 2013
Relevant Coursework: Principles of Computing, and Modern Analytical Instrumentation

TECHNICAL SKILLS

AI-Assisted Research Workflows: LLM-based coding agents for pipeline prototyping and literature triage, prompt-driven pipeline prototyping
Languages: Python, R, Bash, JavaScript, Go
Bioinformatics & Statistics: HyPhy / Datamonkey (comparative sequence analysis, phylogenetics), single-cell RNA-seq analysis, differential gene expression, gene ontology & network analysis, machine learning for genomics
Tools & Infrastructure: Git/GitHub, Google Cloud Platform, reproducible workflow automation, LaTeX
Education & Outreach Tools: Galaxy Training Network, retrieval-augmented generative-AI e-tutorial development, IRB-approved user-study design

PUBLICATION

Peer-Reviewed Journal Articles

[9] **Kim, H.**, Scheffler, K., Nekrutenko, A., Martin, D. P., Weaver, S., Murrell, B., Kosakovsky Pond, S. L. (2026). The Physicochemical Basis of Protein Evolution: Property-Informed Evolutionary Models (PRIME). *Molecular Biology and Evolution*. doi:10.1093/molbev/msag200 [Preprint | Code | Web App | Data]

[8] Huzar, J., **Kim, H.**, Kumar, S., Miura, S. (2022). MOCA for integrated analysis of gene expression and genetic variation in single cells. *Frontiers in Genetics*, 13:831040. doi:10.3389/fgene.2022.831040

[7] Ding, Y., **Kim, H.**, Madden, K., Loftus, J., Chen, G., Allen, D., Zhang, R., Xu, J., Chen, C., Xu, Y., Tasian, S., Tan, K. (2021). Network Analysis Reveals Synergistic Genetic Dependencies for Rational Combination Therapy in

Philadelphia Chromosome-like Acute Lymphoblastic Leukemia. *Clinical Cancer Research*, 27(18). doi:10.1158/1078-0432.CCR-21-0553

[6] Tarca, A. L., Pataki, B. Á., Romero, R., Sirota, M., Guan, Y., Kutum, R., Gomez-Lopez, N., Done, B., Bhatti, G., Yu, T., Andreoletti, G., Chaiworapongsa, T., **The DREAM Preterm Birth Prediction Challenge Consortium**, Hassan, S. S., Hsu, C., Aghaeepour, N., Stolovitzky, G., Csabai, I., Costello, J. C. (2021). Crowdsourcing assessment of maternal blood multi-omics for predicting gestational age and preterm birth. *Cell Reports Medicine*, 2(6). doi:10.1016/j.xcrm.2021.100323

[5] Ichikawa, Y., Bruno, V. M., Woolford, C. A., **Kim, H.**, Do, E., Brewer, G., Mitchell, A. P. (2021). Environmentally contingent control of Candida albicans cell wall integrity by transcriptional regulator Cup9. *Genetics*, 218 (3). doi:10.1093/genetics/iyab075

[4] Tao, Y., Rajaraman, A., Cui, X., Cui, Z., Chen, H., Zhao, Y., Eaton, J., **Kim, H.**, Ma, J., Schwartz, R. (2021). Assessing the Contribution of Tumor Mutational Phenotypes to Cancer Progression Risk. *PLOS Computational Biology*, 17(3). doi:10.1371/journal.pcbi.1008777

[3] He, B., Gao, P., Ding, Y., Chen, C., Chen, G., Chen, C., **Kim, H.**, Tasian, S. K., Hunger, S. P., Tan, K. (2020). Diverse noncoding mutations contribute to deregulation of cis-regulatory landscape in pediatric cancers. *Science Advances*, 6(30). doi:10.1126/sciadv.aba3064

[2] Lin, C., Jain, S., **Kim, H.**, Bar-Joseph, Z. (2017). Using neural networks for reducing the dimensions of single-cell RNA-Seq data. *Nucleic Acids Research*, 45(17). doi:10.1093/nar/gkx681

[1] Kim, H. (2023). Running a summer journal club for an interdisciplinary community: how to maintain engagement when members have disparate prior knowledge. *The Journal for Research and Practice in College Teaching*, 8(2). https://journals.uc.edu/index.php/jrpct/article/view/8150/6710

Peer-Reviewed Conference & Workshop Papers

[2] Kim, H., Kosakovsky Pond, S. L., MacNeil, S. (2025). Conversations over Clicks: Impact of Chatbots on Information Search in Interdisciplinary Learning. *2025 IEEE Frontiers in Education Conference (FIE)*. doi:10.1109/FIE63693.2025.11328556
- FIE 2025 [Preprint | Teasers | Slide | Video | Recap]

[1] Kim, H., Kosakovsky Pond, S. L., MacNeil, S. (2024). WIP: Identifying Tutorial Affordances for Interdisciplinary Learning Environments. *2024 IEEE Frontiers in Education Conference (FIE)*. doi:10.1109/FIE61694.2024.10893187
- FIE 2024 [Preprint | Slide | Video | Recap]

Preprints

[4] Kim, H. (2026). Entering a New Domain in the Era of Generative AI: An Autoethnography of Solo Game Development as a Bioinformatics Educator. *arXiv* preprint arXiv:2611.09623. doi: 10.48550/arXiv.2611.09623. [Data]

[3] Kim, H., Nabid, R. A., Sorathiya, J., Doan, M., Jordan, E., Nasimova, R., Kosakovsky Pond, S. L., MacNeil, S. (2026). Changing the Optics: Comparing Traditional and Retrieval-Augmented GenAI E-Tutorials in Interdisciplinary Learning. *arXiv* preprint arXiv:2602.20544. doi:10.48550/arXiv.2602.20544 [Preprint page | Task | Trad E-Tutorial | RAG E-Tutorial]

[2] Jiomekong, A., McGinty, H. K., Mills, K. G., Oelen, A., Rajabi, E., McElroy, H., Christou, A., Saini, A., Zebaze, J. A., **Kim, H.**, Jacyszyn, A. M. (2025). Charting the future of scholarly knowledge with AI: A community perspective. *arXiv* preprint arXiv:2509.02581. doi:10.48550/arXiv.2509.02581

[1] Tao, Y., Rajaraman, A., Cui, X., Cui, Z., Eaton, J., **Kim, H.**, Ma, J., Schwartz, R. (2019). Improving personalized prediction of cancer prognoses with clonal evolution models. *bioRxiv*. doi:10.1101/761510

Editorship

[1] Auer, S., Betz, D., Biniossek, C., Jacyszyn, A., Jiomekong, A., **Kim, H.**, McGinty, H.K., Mills, K.G., Oelen, A., Rabby, G. and Rajabi, E. (2026, eds.). Proceedings of the 2nd AAAI Bridge on Artificial Intelligence for Scholarly Communication. *Open Conference Proceedings*, 8. doi:10.52825/ocp.v8i.3212

POSTERS

Ahmed, M., Kashyap, D., **Kim, H.**, Duy, C.* (2025, April 25-30). *DNMT1 promotes escape from chemotherapy-induced senescence in acute myeloid leukemia* [Abstract]. In Proceedings of the 116th Annual Meeting of the American Association for Cancer Research. Chicago, IL: AACR. Abstract nr 0293 / 2.

Kim, H.* (2024). *It Takes a Village: Understanding the Community Aspects of Interdisciplinary Tutorials* Poster: 2024 ASEE/IEEE Frontiers in Education Conference, Washington, D.C.

Kim, H.*, Kosakovsky Pond, S. L. (2023). *PRSuite: PRoperty Informed Models of Evolution (PRIME), the Imputation (PREI), and the Visualization (PReC).* Poster: The Society for Molecular Biology and Evolution 23 Conference, Ferrara, Italy.

Kim, H.*, Shank, S., Kosakovsky Pond, S. L. (2022). *PRoperty Informed Models of Evolution (PRIME).* Poster: The 31st KSEA Northeast Regional Conference, Virtual.

Hu, Y., Chen, C., Ding, Y.*, **Kim, H.**, Tan, K. (2019). *Synergistic Control Genes in Cancer Gene Networks as Targets for Combination Therapy.* Poster: Children's Hospital of Philadelphia Research Poster day and Scientific Symposium, Philadelphia, PA.

INVITED TALKS

Kim, H.* (2026). *Informal Learning in the Era of Generative AI*. Seminar: Philadelphia Developer Group, Virtual.

Kim, H.* (2026). *New Affordances and Limitations of E-Tutorials in the Era of Generative AI.* Invited Talk: Penn State Berks AI Week (April 14-17), Reading, PA. [Flyer | Recap]

Kim, H.* (2024). *Characterization of Disciplinary Hybridization in Bioinformatics Software Tutorials.* Seminar: Philadelphia Korean Scholars Association, Philadelphia, PA.

Kim, H.* (2023). *Using HyPhy package on comparative sequence analysis.* Workshop: Temple Bioinformatics Studio, Philadelphia, PA. [Flyer]

Kim, H.* (2023). *Data Analysis in Bioinformatics Research (in academia).* Seminar: Philadelphia Developer Group, Virtual.

Kim, H.*, Kosakovsky Pond, S. L. (2022). *PRIME Evolutionary Imputation (PREI).* Flash Talk: International Conference on Intelligent Biology and Medicine, Philadelphia, PA.

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate Sep 2026 - present
University of Mississippi – Sayaka Miura lab *Remote*

- Reconstructing cancer phylogeny at single-cell resolution by integrating evolutionary models with single-cell multi-omics data.

Project Lead Feb 2024 - May 2026
Temple University ACME Group & Temple University HCI lab *Philadelphia, PA*

- Identified gaps in knowledge and proposed ideas that can make data and method work together.
- Reconciled different research interests of stakeholders involved.

- Designed a user study, drafted a protocol approved by the institutional review board, and ran the study whose results we reported in *Conversations over Clicks* (Kim et al., 2025).
- Team: Saumika Pandey (09/2025-04/2026 & 01/2025-05/2025); Hena Patel (05/2025-04/2026); Rayhana Nasimova (05/2025-04/2026); Elijah Jordan (05/2025-04/2026); Jeni Sorathiya (01/2025-04/2026); Rahad Arman Nabid (01/2025-04/2026); Kriti Baru (01/2025-05/2025); Minh Doan (01/2025-05/2025)

Bioinformatics Engineer Oct 2021 - Jan 2022
Lifetime Omics *Remote*

- Automated cutting-edge methods and analyzed COVID-19 metagenomics data in the Google Cloud environment.
- Developed detailed standard operating procedure for reproducibility.
- Took initiatives to solve problems and demonstrated dedication for the project in the startup environment.

Bioinformatics Analyst/Software Engineer Jul 2017 - Jun 2019
Children's Hospital of Philadelphia – Kai Tan lab *Philadelphia, PA*

- Investigated RNA-Seq and microarray data from B-Cell Acute Lymphoblastic Leukemia subtypes using differential gene expression analysis, Gene Ontology enrichment analysis, and other relevant bioinformatics methods.
- Identified cancer-specific genetic interactions that led to publications in high-impact journals.
- Facilitated communication in the interdisciplinary environment of doctors and wet-lab and dry-lab researchers.

Research Programmer Feb 2017 - Jun 2017
Carnegie Mellon University – Russell Schwartz lab *Pittsburgh, PA*

- Investigated clinical and genomic data to create a cancer progression analysis pipeline using machine learning.

MS Graduate Researcher Jan 2016 - Dec 2016
Carnegie Mellon University – Ziv Bar-Joseph lab *Pittsburgh, PA*

- Curated and analyzed single-cell gene expression data.

Post-Baccalaureate Researcher (Chemistry/Biology) Aug 2013 - Jun 2014
Carnegie Mellon University – Drs. Fred Lanni, Aaron Mitchell, and Luisa Hiller *Pittsburgh, PA*

- Tested various polymer surface coatings for the prevention and destruction of biofilms using red/NIR light.

Undergraduate Student Researcher (Chemistry) Jan 2012 - Aug 2013
Carnegie Mellon University – Kevin Noonan lab *Pittsburgh, PA*

- Characterized pyrylium and pyridinium salts by modifying functional groups.

Student Researcher (Biology) Jun 2011 - Aug 2011
Carnegie Mellon University – 2011 Summer Research Institute *Pittsburgh, PA*

- Analyzed interactions among three ribosomal assembly factors in *Saccharomyces cerevisiae*.
- 1 of 12 student researchers selected for the program.

AWARDS, FELLOWSHIPS, & GRANTS

2023 Best Poster Award, 2nd Place (Computer and Information Sciences), The 36th US-Korea Conference

2022 & 2023 Selected Attendee Support, Scientists and Engineers Early Career Development Workshop

2022 KSEA Excellent Poster Award, The 31st KSEA Northeast Regional Conference

2022 CST Three-Minute Thesis Competition 2nd Place Award, Temple University (College-level)

2015 Departmental Merit Fellowship, Carnegie Mellon University

2013 Mellon College of Science Research Honors, Carnegie Mellon University

2012 Summer Undergraduate Research Fellowship, Carnegie Mellon University

TEACHING EXPERIENCE

Client Jan 2025 - Apr 2025
Temple University *Projects In Computer Science (CIS-4398)*

- Provided feedback to a project as a client in Professor Ian Tyler Applebaum's senior capstone class for 6 weeks [Class Website | Team Documentation Website | Pitch as a Client | Project Final Demo].
- Team size: 7.

Co-Coordinator/Reflective Practicum Completion for THE Certificate May 2023 - Jul 2023
Temple University *Deep Learning for the Life Sciences Journal Club*

- Co-organized a 10-week summer journal club with Dr. Enzo Carnevale for researchers of diverse levels of expertise (undergraduate students, graduate students, postdocs, alumni, visitors, and professors); wrote up the lessons learned on maintaining engagement across disparate prior knowledge in *The Journal for Research and Practice in College Teaching* (2023).
- Built a unique inter-departmental community in the College of Science and Technology.
- Discussed teaching methods and reflections with Dr. Jay Lunden (reflective practicum mentor) for 7 weeks.
- Group size: 35.

Course Completion for Teaching in Higher Education (THE) Certificate Aug 2022 - Dec 2022
Temple University *Teaching in Higher Education (EPSY-8985)*

- Developed syllabi, assignments, and assessments using the principles of integrated course design.
- Applied the current theories of teaching in different contexts.
- Used a variety of effective teaching methods to address learners universally.
- Discussed a reflective and purposeful approach to teaching with other instructors.

Teaching Assistant Aug 2020 - Dec 2020
Temple University *Genomics in Medicine (BIOL-3111/5111)*

- Generated formative and summative assessment materials and provided timely feedback (Youtube/Introduction).
- Class size: 150.

Teaching Assistant Aug 2019 - May 2020
Temple University *Wet Lab Courses (BIOL-2112 and BIOL-1012)*

- Gave a short lecture in the beginning of every lab, monitored student performance, and provided guidance.
- Graded lab reports and generated quizzes.
- Class size: 20 (× 2 sections) each.
  - BIOL-1012 General Biology II, Spring 2020 was an introductory wet lab course for non-biology majors.
  - BIOL-2112 Introduction to Cellular and Molecular Biology, Fall 2019 was a lab for biology majors.

Course Developer Feb 2016 - Aug 2016
Carnegie Mellon University *Programming for Scientists (02201/02601)*

- Generated open-source course materials (codes and instructions) with Drs. Phillip Compeau and Carl Kingsford for Go-lang beginners.

TUTORIAL

[2] Kim, H. [Hannah Kim in Comp Bio]. (2025, May 28). *#summer2025 #hci [Research Project Teaser] Bioinformatics Software Tutorial Supporting Chatbot* [Video]. YouTube. https://youtu.be/cfTl6e58DPU
[1] Kim, H. [GalaxyProject]. (2024, Sep 13). *GTN Tutorial: Chloroplast Genome Assembly* [Video]. YouTube. https://youtu.be/gIcbY9kXdTo?si=FrJLLLb0aWNOk-Ck

SERVICE & LEADERSHIP

Education Standards Committee Member Nov 2025 - present
P3722 IEEE Standard Guide for Responsible AI in Higher Education

- As a member of the working group P3722, participated in monthly meetings to discuss and advance the development of the standard.
- As a member of the subgroups (1) Overview and Guiding Principles, (2) Educator Development (specifically Institutional Support), and (3) External Awareness (specifically Risk Management), discussed the purpose and scope of the guideline, identified areas for further study, and drafted sections with future-proofing in mind.

AI for Scholarly Communication (AI4SC) Bridge Organizing Committee Sep 2025 - present

- **Member for 3rd AI4SC, AAAI-27** Sep 2026 - present
- **Member for 2nd AI4SC, AAAI-26** Sep 2025 - Jan 2026
  - Contributed to the program document drafts, generated recruitment materials, and attended bi-weekly meetings.
  - Co-managed bridge submissions as an editor on the TIB Open Conference Proceedings platform.

College of Science and Technology Alumni Board Member Jun 2026 - present
Temple University College of Science and Technology Alumni Board

- Part of (1) Careers and Networking and (2) Events committees.

Alumni Network Leader Apr 2025 - present
Carnegie Mellon University Philadelphia Network

- Hosted local events (Globally Plaid Recap, Sep 2025) for the alumni and affiliates.

Korean-American Innovative Technology Engineers and Entrepreneurs (KITEE) Jan 2023 - Aug 2026

- **Young Generation Director I (PhD level)** Aug 2025 - Aug 2026
  - Supported organizing KITEE events and promoting membership drive.
- **The 33rd KSEA NRC Steering Committee Member & PR Director** Jan 2024 - Apr 2024
  - Planned the conference in the weekly committee meeting with PIs and staff from academia and industry.
  - Discussed conference planning with other members; an emphasis on diversity and inclusion (i.e. accessibility & childcare support).
  - Supported onsite facilitation on the day of event with >80 attendees.
- **Event Director** Oct 2023 - Dec 2024
  - Connected with graduate students in the northeast region in the US to promote KITEE events.
  - Brainstormed ideas during event meetings and supported all registration-related endeavors.
  - Organized KITEE Entrepreneurship Forum 2023 (>80 attendees) as part of the committee.
- **The 2023 KITEE-FELIX Hackathon Committee Secretary** Feb 2023 - Apr 2023
  - Organized the hackathon in the weekly committee meeting.

- Reached out to universities and engineer communities in the Philadelphia region.
- Supported the onsite activities.

- **The 32nd KSEA NRC Committee Philadelphia Young Generation Director** Jan 2023 - Apr 2023
  - Planned the Northeast Regional Conference (NRC) conference in the bi-weekly committee meeting with PIs and staff from academia and industry.
  - Operated as the public relations and promotion chief for the pre-networking event before the conference.
  - Promoted networking (e.g. networking bingo) and facilitated the event flow onsite (104 registrants).

University Service (Temple University)

- **Panelist** Oct 2024 - Aug 2025
  - CST Graduate Orientation, 08/2025, Graduate Admissions Team in College of Science and Technology: shared graduate school tips and tricks with the new graduate students.
  - Publishing Journal Articles: A Panel Discussion, 10/2024, Temple University Graduate School: served as one of four panelists and shared solo- and first-author experiences with graduate students across the community.
- **CST-Graduate Student Organization Board Member** Dec 2021 - May 2024
  - **Media Chair** (May 2023 - May 2024): advertised organization events on social media; organized Graduate Student Fall and Spring social events, each with 30+ attendees.
  - **DEI Representative** (Dec 2021 - Apr 2023): identified potential DEI topics and suggested strategies; facilitated the success of the yearly PhD/MS alumni panelist event involving all six departments within the college for two years in a row.
- **Student Representative, CST Diversity, Equity, and Inclusion (CST-DEI) Committee** Nov 2021 - Aug 2023
  - Discussed the promotion of DEI within the college at the bi-weekly committee meetings; provided feedback for a wide array of DEI topics.
- **Vice President, Biology Graduate Student Society (BGSS)** Sep 2021 - Sep 2023
  - Facilitated communication between the department and the graduate school; evaluated travel grant applications; led action plans to address diverse career needs; generated databases for student placements after graduation; supported on-boarding of first-year students; planned and oversaw activities for the Fall 2022 departmental retreat.

PEER REVIEW & COMMUNITY INVOLVEMENT

Ad Hoc Reviewer

- [Journal, 1×] Frontiers in Education (2026 IF=2.6), *10/2026*
- [Journal, 1×] Frontiers in Antibiotics (2026 IF=3.9), *09/2026*
- [Workshop, 1×] Second International Workshop on Users and Knowledge Graphs as part of 22nd International Conference on Semantic Systems, *07/2026*
- [Journal, 1×] JMIR Medical Informatics (2025 IF=5.0), *07/2026*
- [Workshop, 2×] KG4Health Workshop Program Committee as part of 24th International Conference on Artificial Intelligence in Medicine, *05/2026*
- [Conference, 2×] 2026 IEEE Frontiers in Education Conference (FIE), *04/2026*
- [Journal, 1×] Virus Evolution (2024 IF=4.0), *01/2026*

- [Conference, 2×] 2026 IEEE Global Engineering Education Conference (EDUCON), *12/2025*
- [Journal, 1×] Virus Evolution (2024 IF=4.0), *11/2025*
- [Conference, 2×] 2025 IEEE FIE, *04/2025*
- [Conference, 3×] 2025 IEEE EDUCON, *11/2024*
- [Conference, 2×] 2024 IEEE FIE, *07/2024*
- [Organization] Temple University CST Diversity Innovation Initiative Fund, *03/2024*
- [Journal, 1×] Scientific Reports (2021 IF=4.996), *04/2023*
- [Journal, 1×] Frontiers in Oncology (2021 IF=5.738), *06/2022*

Science Fair Judge

- [5×] George Washington Carver Science Fair, *03/2026, 03/2025, 04/2024, 03/2023 & 03/2022*
  - Judged science fair projects done by students in grades 4 and 5 (04/2024) and grades 6 to 12 (03/2026, 03/2025, 03/2022 & 03/2023) with a group of educators.
  - Judged Behavior and Social Sciences category in 03/2026 and 03/2025.

PROFESSIONAL MEMBERSHIPS

- **Knowledge Knocking for Your Universe (KKYU),** *05/2025-present*
- **PhillyCHI,** *01/2025-present*
- **Galaxy Training Network Contributor,** *07/2024-present*
- **Korean-American Women in Science and Engineering (KWiSE),** *08/2023-present*
- **Philadelphia Developer Group,** *04/2023-present*
- **Korean-American Innovative Technology Engineers and Entrepreneurs (KITEE),** *04/2022-present*
- **AnitaB.org**, *09/2021-present*
- **Korean-American Scientists and Engineers Association (KSEA)**, *02/2020-present*
- **Philadelphia Korean Scholars Association (PKSA)**, *06/2019-present*
- Past: Temple AAPI Faculty & Staff Affinity Group (02/2024-05/2026); KASBP (10/2022-07/2024); research collaborations with Fox Chase Cancer Center and Temple iGEM (12/2022-11/2024)